

Game Development Project

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Game Information

Game Title: DEEP SEA DIVER

Genre: Endless Survival / Action

Game Description

Deep Sea Diver is an immersive underwater survival game that challenges players to embody a professional diver exploring the vast and dangerous depths of the ocean. The primary objective is to descend as far as possible, navigating a treacherous environment while managing a critically limited oxygen supply. To aid in their journey, players can collect gold from sunken treasure chests. This gold fuels a persistent upgrade system, allowing for the enhancement of equipment across multiple playthroughs, creating a compelling loop of progression and repeated attempts to conquer the abyss.

Game Play Mechanics

The core gameplay revolves around precise control and resource management. The player maneuvers the diver using standard keyboard inputs (WASD). As the diver descends, their oxygen level depletes at a constant rate, creating a persistent sense of urgency. To survive, players must strategically locate and collect oxygen tanks scattered throughout the environment. The ocean is fraught with hazards, most notably jellyfish, which upon contact, cause significant damage to the diver's oxygen supply. Players can mitigate this threat by

collecting temporary shields for protection. The game's physics, including movement, gravity, and collision detection, are robustly handled by the Phaser 3 Arcade Physics engine, ensuring a responsive and believable underwater experience.

Player Engagement & Target Audience

Motivations for Playing

The design of Deep Sea Diver is centered around three key motivational pillars to ensure a compelling and replayable experience:

1. > **High Score Chasing:** The endless nature of the descent encourages players to constantly challenge themselves and their friends, striving to beat their previous maximum depth record and achieve mastery over the game's mechanics.
2. > **Meaningful Progression:** The in-game shop system provides a tangible sense of growth and accomplishment. Gold collected in each run can be invested in permanent upgrades, making subsequent dives more manageable and empowering the player over time.
3. > **Intense Challenge:** The core loop of managing a constantly diminishing resource (oxygen) while simultaneously dodging dynamic obstacles creates a high-stakes, adrenaline-fueled environment that rewards focus, quick reflexes, and strategic planning.

Targeted Audience

The game is primarily designed for two overlapping player demographics. Firstly, it appeals to casual gamers who appreciate accessible, arcade-style challenges that are easy to learn but difficult to master. Secondly, it targets players who enjoy 'rogue-lite' mechanics, where performance and collection in one run directly contribute to long-term success and character progression, fostering a "one more run" mentality.



Technical Requirements

Achievement

The project successfully implemented a wide range of technical features using the Phaser 3 framework to create a complete and polished game experience. The key achievements are detailed below:

- > **Scenes:** Full implementation of a multi-scene architecture, including a main **MenuScene** for game start, the core **PlayScene** for gameplay, and a **ShopScene** for player upgrades.
- > **Background Images:** Aesthetically pleasing, multi-layered ocean backgrounds were created to establish a deep-sea atmosphere that darkens as the player descends.
- > **Object Manipulation:** Dynamic and procedural spawning of essential game objects, including treasure chests, oxygen tanks, and environmental hazards like jellyfish, ensuring unique gameplay in every run.
- > **Spritesheets:** Use of spritesheets for fluid character animation, including diver movements and visual effects for the protective shield.
- > **Physics:** Comprehensive implementation of the Arcade Physics engine, managing gravity, player velocity, world bounds, and all collision and overlap detection between the player and other game objects.
- > **Input:** Versatile input support for Keyboard (both WASD and Arrow keys), Mouse clicks, and Gamepads, ensuring broad accessibility.
- > **Cameras:** A smooth follow-camera logic was implemented to keep the diver centered on the screen during their descent, providing a consistent player perspective.
- > **Tile Sprites:** Parallax scrolling layers of fish and other sea life were implemented using tile sprites to create a convincing illusion of environmental depth and movement.
- > **Audio:** Integration of sound effects for key game events, including item collection, taking damage, and user interface interactions, enhancing player feedback and immersion.



Instruction Manual

How to Run the Game

To play Deep Sea Diver, please follow these simple steps:

- 1 Extract the contents of the provided ZIP file to a folder on your computer.
- 2 The game is a web-based project and requires a local server to run correctly. Use a tool like the "Live Server" extension in Visual Studio Code to open the `index.html` file.
- 3 The game will launch in your default web browser.

Game Controls & Notes

Movement: Use **WASD** or the **Arrow Keys** to control the diver.

Boost: Hold down the **Shift** key to use a temporary speed boost.

Pause: Press the **ESC** key to pause and un-pause the game.

Important Note: The game utilizes your browser's **LocalStorage** to save your total gold and purchased upgrades. Your progress will be automatically saved between sessions on the same browser.